built once from the fault-free period (Events: also daily)

## 1–2 SENSE the building's own data, turned into events

### Trend data (1)
What the building-automation system already records: temperatures, flows, damper and valve positions, fan speeds, once a minute.

### Configuration file (1)
Maps sensor names to roles and lists the control sequence as rules. The only building-specific part: a new building means a new file, not code.

### Events (2)
**State events: what the equipment is doing**

**Signature events: what should not happen**

One case per day. Rebuilt every new day.

*state events only, into model discovery (3)*

## 3–4a UNDERSTAND what a normal day looks like, learned from one fault-free period

### Process models (3)
Discovered from the state events only, so they cannot learn the configured fault rules. One per terminal unit; one for the whole unit.

### Calibration (4a)
Healthy bands and model thresholds for every check come from the fault-free days. No threshold ever sees fault data.

### Held-out days (4a)
The last eight days of each month (96 a year) are kept aside and used only to measure each check's false-alarm rate.

and feed the Rules check only **signature events skip discovery**

*the discovered models and the calibrated thresholds*

run on every new day

## 4b CHECK eight checks score every new day, each against its own threshold

| Rules | Residual | Oscillation | Unit conformance | Terminal-unit conformance | Absence | Frequency | Rate |
|---|---|---|---|---|---|---|---|
| control-sequence rule broken | physical relationship drifted | actuator reversing too often | day fits the unit model | day fits the terminal-unit model | expected terminal unit silent | daily count out of range | last 30 days against same month; support only |

*read the raw sensors* — *read the event log*

*each check's flagged days*

## 5–6 ALERT only what beats its own false-alarm rate reaches the operator

### Noise gate (5)
A check may alarm only when its flags exceed what its own false-alarm rate would produce by chance. The rate is never taken below 3 days in 96.

### Alarm to the operator (6)
What broke. Where: unit, terminal unit, zone.
The evidence: values and duration.
The check's measured false-alarm rate.

**STRATA**
a process-mining framework for HVAC fault detection